



SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL
SCIENCES
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



COURSE NAME: Theory of Computation

COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO1: Apply mathematical foundations such as formal proofs, induction, and basic concepts of automata theory to solve computational problems.(BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5
Duration : 1 Hour

Total Marks:50

Analytical Problem

Code of Conduct:

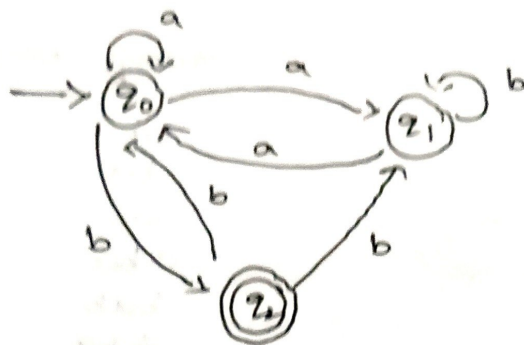
I N Ramana (192372228) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

N Ramana

Signature of the student:

(1) Construct DFA from the given NFA



Sol:

States $(Q) = \{q_0, q_1, q_2\}$

Alphabet $(\Sigma) = \{a, b\}$

Start state : q_0

Final state : q_2

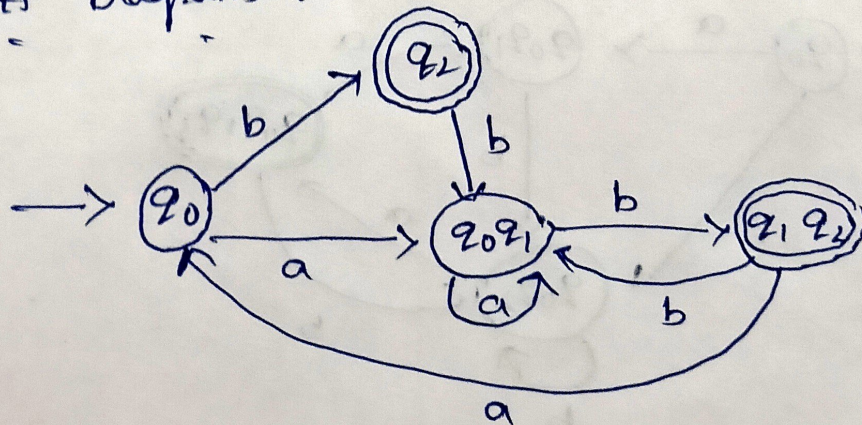
NFA Transition table

state	a	b
$\rightarrow q_0$	$q_0 q_1$	q_2
q_1	q_0	q_1
$* q_2$	$\emptyset$	$q_0 q_1$

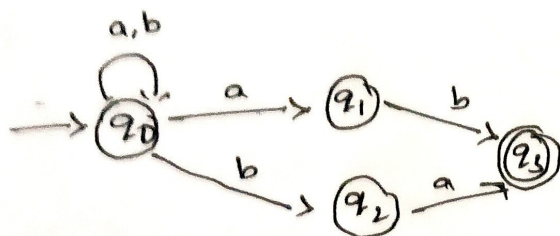
DFA Transition table

state	a	b
$\rightarrow q_0$	$q_0 q_1$	q_2
$q_0 q_1$	$q_0 q_1$	$q_2 q_1$
q_2	$\emptyset$	$q_0 q_1$
$q_2 q_1$	q_0	$q_0 q_1$

DFA Diagram :-



(2) construct a DFA from a NFA that ends with ab or ba



strings accept

ab
ba
aab
aba
bab
bba

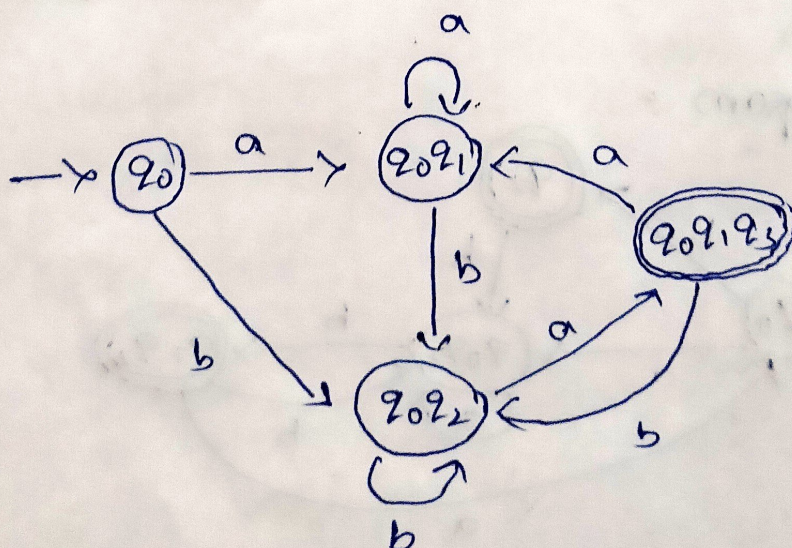
NFA Transition table

state	a	b
$\rightarrow q_0$	$q_0 q_1$	$q_0 q_2$
q_1	$\emptyset$	q_3
q_2	q_3	$\emptyset$
q_3	$\emptyset$	$\emptyset$

DFA Transition Table

	a	b
$\rightarrow q_0$	$q_0 q_1$	$q_0 q_2$
$q_0 q_1$	$q_0 q_1$	$q_0 q_2$
$q_0 q_2$	$q_0 q_1 q_3$	$q_0 q_2$
$q_0 q_1 q_3$	$q_0 q_1$	$q_0 q_2$

DFA Diagram



(3A) Given to Design to design a DFA which accepts the string that ends with either 00 (0s), 11

where: $Q = q_0, q_1, q_2, q_3$

$\Sigma = 0, 1$

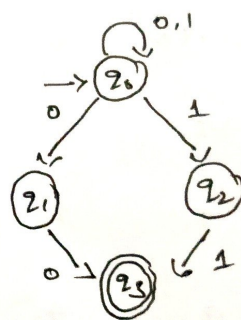
q_0 = start state

q_3 = final state

Transition table

state	0	1
$\rightarrow q_0$	q_0, q_1	q_0, q_1
q_1	q_3	$\emptyset$
q_2	$\emptyset$	q_2
q_3	$\emptyset$	$\emptyset$

DFA Diagram



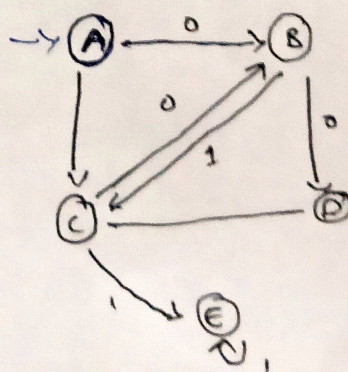
To show 01011 is accepted

input string 01011, start at $q_0 \rightarrow q_0 \rightarrow q_1 \rightarrow q_2 \rightarrow q_3$
it reaches final state then the string accepted

NFA Transition Table

state	0	1
A	B	C
B	D	C
C	B	E
* D	D	C
* E	B	E

NFA Diagram



(4A)

Construct a DFA from the given NFA which accepts 001 as substring

states

$q_0 \rightarrow$ start

$q_1 \rightarrow$ 0

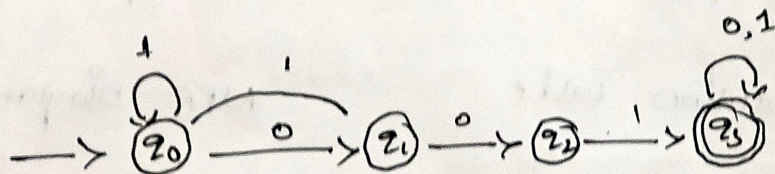
$q_2 \rightarrow$ 00

$q_3 \rightarrow$ 001

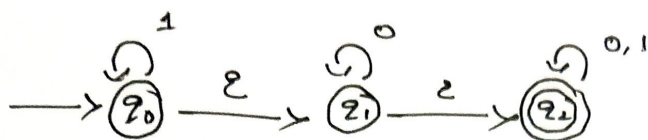
Transition Table

state	0	1
$\rightarrow q_0$	q_1	q_0
q_1	q_2	q_0
q_2	q_3	q_3
$\odot q_3$	q_3	q_3

DFA diagram



(5A) Given a NFA with ϵ -transition, find ϵ -closure and redraw the NFA diagram with start and end states.



start state = q_0

final state = q_1

find ϵ -closure(q_0)

1. $q_0 \rightarrow \epsilon$

$q_1 \rightarrow \epsilon q_2$

ϵ -closure(q_0) = q_0, q_1, q_2

ϵ -closure(q_1) = q_1, q_2

ϵ -closure(q_2) = q_2

DFA states are

$A = \{q_0, q_1, q_2\}$

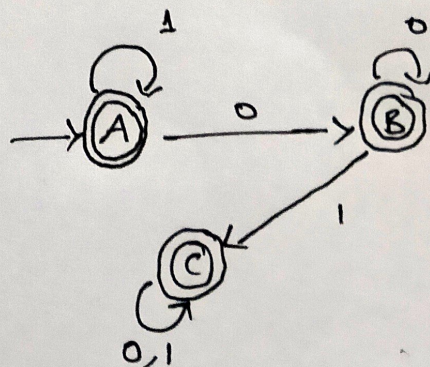
$B = \{q_1, q_2\}$

$C = \{q_2\}$

DFA Transition Table

states	0	1
$\rightarrow A$	B	A
B	B	C
C	C	C

DFA Diagram





SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL
SCIENCES
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



RUBRICS FOR CALCULATION

Criteria	Weightage	Q1 (10)	Q2 (10)	Q3 (10)	Q4 (10)	Q5 (10)	Total Marks (50)
Conceptual Understanding	30	3	3	3	3	3	15
Accuracy of Answer	25	2.5	2.5	2.5	2.5	2.5	12.5
Application of Knowledge	20	2	2	2	2	2	10
Completeness of Response	15	1.5	1.5	1.5	1.5	1.5	7.5
Clarity & Presentation	10	1	1	1	1	1	5
Total per Question	10 Marks	/10	/10	/10	/10	/10	/ 50

